

Self-Service Coffee Kiosk System

Architectural Justification

Architecture Selection:

We chose Layered Architecture for the Self-Service Coffee Kiosk System.

Reasons for Selection:

1. Clear separation of responsibilities: The Presentation Layer handles touchscreen interactions, the Business Layer manages orders, payments, and receipt printing, and the Data Layer stores menu and pricing information.
2. Easy maintenance: Separating the system into layers makes it easier to modify or maintain individual components without affecting the entire system.

Security Advantage:

The layered structure separates the user interface from data storage and payment processing. This helps control access between layers and prevents direct access to the database from the user interface.

Performance Benefit:

The components communicate through defined interfaces, allowing requests such as order processing, payment processing, and database queries to be handled in an organized manner.